

Rectified Flow for Chromosome Detection: Stable Coupling and Few-Step Inference

Author Name, *Member, IEEE* and Co-Author Name, *Senior Member, IEEE*

Abstract—Chromosome karyotyping — the visual inspection of metaphase chromosomes for clinical genetics — requires trained cytogeneticists to classify roughly 46 tightly packed chromosomes per cell into 24 morphologically similar classes, a process that is slow and observer-dependent. Automating this analysis demands a detector that is both accurate and fast enough for clinical deployment, but three obstacles arise: limited accuracy of conventional detectors on fine-grained chromosomes; slow many-step inference and curved-trajectory truncation errors of Denoising Diffusion Probabilistic Model (DDPM)-based diffusion detectors; and training instability on small clinical datasets.

We address these obstacles with *Rectified Flow* (RF), which replaces the stochastic DDPM process with deterministic straight-line ODE paths, instantiated as *KaryoFlow*. The RF training paradigm is the dominant source of accuracy gain: *KaryoFlow* exceeds DiffusionDet by +0.076 mAP and matches Cascade R-CNN and YOLOX-S; a solver×step disentanglement attributes 94% of the gain to RF. We prove an upper bound $\Delta H \leq \log K$ on conditional-entropy reduction from OT coupling in the low-dimensional ($\mathbb{R}^4$) detection space — characterizing *OT Diversity Collapse* — and propose Stochastic Coupling via Sinkhorn transport, which yields a large, highly significant mAP gain in the low-data regime (+0.034, $p < 10^{-120}$) that diminishes with dataset size, and reduces within-run convergence oscillation by 4.6×. DPM-Solver++ for four-step inference achieves a small but statistically significant precision advantage over Heun at matched step count (+0.006 per-image mAP, Wilcoxon $p < 0.001$) with 1.71× speedup, combined with Top- K pruning.

All claims are validated on two public chromosome datasets with multi-seed experiments, per-class AP analysis, and SOTA comparison.

Index Terms—Rectified Flow, object detection, optimal transport, diffusion models, medical image analysis, chromosome karyotyping

I. INTRODUCTION

a) Motivation.: Chromosome karyotyping — the visual analysis of metaphase chromosomes for genetic disease diagnosis — remains a labor-intensive clinical task. Each metaphase image contains about 46 tightly packed chromosomes across 24 classes (A1–Y), with severe class imbalance, fine-grained intra-group similarity, and frequent overlaps. Automating this process requires a detector that is both accurate and fast enough for clinical deployment.

Diffusion models offer a compelling paradigm for detection by framing object localization as iterative denoising from noisy boxes to structured predictions [1]. However, DDPM-based diffusion detectors suffer from slow inference (8–1000 steps)

and curved trajectories that introduce truncation errors in few-step regimes. Rectified Flow [2] replaces the stochastic DDPM process with deterministic straight-line ODE paths, enabling few-step inference — but applying RF to detection raises fundamental questions about coupling design, solver selection, and training stability. We frame these as the *key bottlenecks of RF-based dense detection in low-data regimes*: (i) coupling diversity collapse in low-dimensional structured prediction; (ii) numerical solver design for efficient few-step inference; and (iii) training stability under high object density. Our three contributions address these bottlenecks systematically.

b) From scenario to method.: A typical metaphase spread contains roughly forty-six chromosomes across twenty-four classes, many touching or overlapping. The difficulty is uneven: C-group chromosomes (C6–C12) are morphologically similar, distinguished mainly by subtle banding patterns, while the Y chromosome is among the smallest, appears in only one copy in male samples, and has roughly 1,800 training samples versus about 7,000 per autosome. A useful detector must therefore localize densely packed objects under few-step inference, train stably from an imbalanced small corpus, and run fast enough for interactive screening. These demands map onto our three ingredients: Rectified Flow supplies straight-line trajectories for few-step, low-truncation inference; Stochastic Coupling counteracts OT diversity collapse; and DPM-Solver++ with Top- K pruning converts trajectories into clinical-grade latency.

c) Contributions.: Our first contribution validates the RF training paradigm for chromosome detection. *KaryoFlow* achieves +0.082 mAP over the Euler baseline on 24 Chromosomes Object and +0.017 over DDPM on the original dataset. Because the A0→A1 comparison changes several variables at once (DDPM→RF, Euler→Heun, 1 → 4 steps), a solver×step disentanglement ablation attributes 94% of the gain to the RF paradigm and only 6% to solver and step-count choices; AdaLN-Zero contributes null individually (Appendix B).

Our second contribution is a theoretical characterization of OT coupling’s failure mode in low-dimensional detection space, with a practical remedy. We prove an upper bound $\Delta H \leq \log K$ on the conditional-entropy reduction, empirically validated to within 0.03%, and propose Stochastic Coupling (Sinkhorn-transport sampling instead of argmax), which reduces within-run epoch-level mAP oscillation by 4.6× and yields a large, highly significant mAP gain in the low-data regime (+0.034, $p < 10^{-120}$) that diminishes with dataset size. An ϵ ablation shows $\epsilon < 1$ is harmful while $\epsilon \geq 1$ saturates.

Our third contribution deploys DPM-Solver++ [3] for four-step inference (1.71× speedup over Heun, mAP 0.859±0.003

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The authors are with [Affiliation], [City], [Country] (e-mail: author@example.com).

Code and models: [https://github.com/\[placeholder\]](https://github.com/[placeholder])

over three seeds). A controlled, per-image paired ablation reveals that DPM-Solver++ yields a small but statistically significant precision advantage over Heun at matched step count (+0.006 per-image mAP, Wilcoxon $p < 0.001$, Table VII), refining FlowDet’s [4] conclusion that higher-order solvers perform worse: at matched steps the higher-order solver is in fact slightly *better*, not worse.

d) Novelty boundary.: Relative to FlowDet [4] (CFM with mini-batch OT, reports higher-order solvers perform worse), our novelty is: (i) the theoretical characterization of *why* mini-batch OT becomes a liability in low-dimensional structured prediction (Section III-C); and (ii) a solver $\times$ step disentanglement showing that, at matched step count, the higher-order DPM-Solver++ is in fact slightly but significantly *better* than Heun (not strictly worse as FlowDet reports), while being $1.71\times$ faster in NFE. Relative to DeFloMat [5] (RF for medical detection, coupling as implementation detail), we provide the theoretical analysis of coupling design as a training pathology and the Stochastic Coupling remedy. Unlike OT-CFM [6] and multisample flow matching [7] (high-dimensional image generation, $d \sim 10^5$), our setting is low-dimensional ($d = 4$) with $K \approx 46$, where OT collapse is severe ($\Delta H/H \approx 0.69$). AdaLN-Zero [8] is reused as a standard implementation detail (Appendix B); DPM-Solver++ is adopted off-the-shelf, our contribution being the controlled disentanglement analysis with per-image paired statistical tests.

These contributions are backed by comprehensive validation. All claims are validated on both Chromosome20240904 (henceforth the *original* dataset, 1,540 images) and 24 Chromosomes Object (5,000 images), with multi-seed coupling ablation, per-class AP analysis, test-set evaluation, and an FPS benchmark.

Figure 1 summarizes the framework.

II. RELATED WORK

a) Diffusion-based object detection.: DiffusionDet [1] formulates detection as iterative denoising from noisy boxes using DDPM, but curved DDPM trajectories make few-step inference slow and truncation-error-prone. DiffuBox [9] extends the diffusion paradigm to 3D object detection via point-diffusion refinement of coarse proposals, but the 3D point-diffusion setting differs substantially from our 2D noise-box-to-GT formulation. FlowDet [4] applies Conditional Flow Matching with mini-batch OT coupling and reports higher-order solvers perform worse, but does not analyze *why* mini-batch OT becomes a liability in low-dimensional structured prediction. DeFloMat [5] uses Rectified Flow for medical detection but treats coupling as an implementation detail. Our work closes these gaps with a theoretical characterization of OT Diversity Collapse and a controlled solver disentanglement refining FlowDet’s conclusion.

b) Rectified Flow and Flow Matching.: Rectified Flow [2] replaces curved DDPM trajectories with straight-line ODE paths, and Flow Matching [10] provides a unified training framework. OT-CFM [6] and multisample flow matching [7] use mini-batch OT coupling successfully in *image*

generation ($d \sim 10^5$, $K \approx$ batch size), but the behavior in low-dimensional structured prediction (d small, K targets per image) has not been characterized in terms of diversity collapse. Concurrent work analyzes OT degradation in conditional high-dimensional generation [11], but the failure mode there (condition-skewed prior) is orthogonal to the low-dimensional diversity collapse we characterize. With $d = 4$ and $K \approx 46$, OT coupling approaches its $\log K$ entropy-reduction upper bound, collapsing coupling diversity. We characterize this failure mode and propose Stochastic Coupling as a remedy interpolating between hard OT and random pairing.

c) Chromosome detection.: Prior work uses conventional detectors (YOLO, Faster R-CNN [12]) that leave a measurable accuracy gap on the fine-grained 24-class setting. ChromosomeNet [13] uses the same Taichung dataset as our 24 Chromosomes Object benchmark but releases no code or pretrained models. We compare against standard detectors with public implementations (Cascade R-CNN, YOLOX-S, DiffusionDet) and provide the first open-source diffusion-based detector for chromosome karyotyping (code will be released upon publication) with multi-seed validation and per-class AP analysis.

III. METHOD

A. Rectified Flow for Detection (KaryoFlow)

a) RF formulation.: The conditional probability path is

$$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \mathbf{x}_1, \quad t \in [0, 1],$$

where x_0 is the target (GT bbox) and x_1 is the source (Gaussian noise). The corresponding velocity field is constant along the path, $v = x_1 - x_0$, and the training objective is the flow matching loss

$$\mathcal{L}_{FM} = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[\left\| \mathbf{v}_\theta(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{x}_0) \right\|^2 \right].$$

Detection-specific adaptation: source $\mathbf{x}_1 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_4)$, target x_0 are GT bboxes, $d = 4$ (vs. $d = 196,608$ in image generation), and $K \approx 46$ targets per image.

b) Time conditioning.: AdaLN-Zero [8] serves as the time conditioning mechanism, replacing scale-shift conditioning with zero-initialized adaptive layer norm so the network starts as an unconditional model (identity at $t=0$). A separate ablation (Appendix B) confirms its individual contribution to mAP is null; the +0.082 mAP gain is entirely attributable to the RF formulation itself, with the shifted noise schedule contributing negligibly on its own (Appendix C). We retain AdaLN-Zero as a standard implementation detail, not a separate contribution.

B. ODE Solvers: Heun and DPM-Solver++

a) Heun solver (2nd-order).: Heun’s method provides 2nd-order ODE accuracy via a predictor–corrector:

$$\begin{aligned} \hat{\mathbf{x}}_{t-\Delta t} &= \mathbf{x}_t + \Delta t \cdot \mathbf{v}_\theta(\mathbf{x}_t, t), \\ \mathbf{x}_{t-\Delta t} &= \mathbf{x}_t + \frac{\Delta t}{2} [\mathbf{v}_\theta(\mathbf{x}_t, t) + \mathbf{v}_\theta(\hat{\mathbf{x}}_{t-\Delta t}, t - \Delta t)]. \end{aligned}$$

Cost: 2 network forward evaluations (NFE) per step. At 4 steps this yields 8 NFE (7 in practice; the last step degrades to Euler).

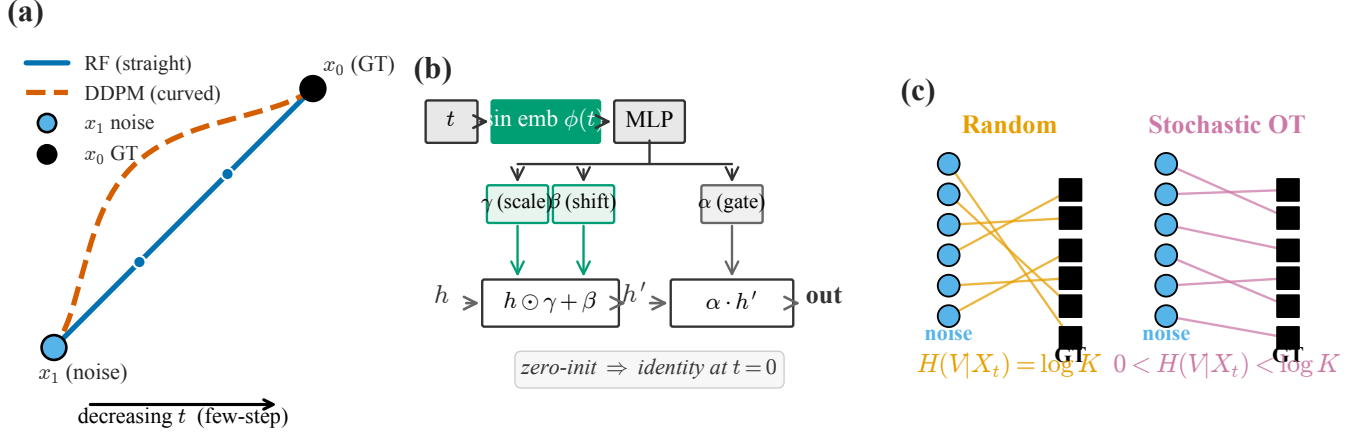


Fig. 1. KaryoFlow overview. (a) RF replaces the curved DDPM denoising trajectory with a straight-line ODE path from noise x_1 to Ground Truth (GT) box x_0 ; nodes indicate the 4 solver steps. (b) Time conditioning injects the continuous time t via AdaLN-Zero zero-initialized modulation so the network is identity at $t=0$. (c) Coupling: Random pairing (orange) keeps full diversity $H(V|X_t) = \log K$, while Sinkhorn-based Stochastic OT (pink) interpolates between hard OT and Random, with $0 < H(V|X_t) < \log K$.

b) DPM-Solver++ (2nd-order multistep, 1 NFE/step):

We adapt DPM-Solver++ [3] to the RF linear path. The model directly predicts x_0 (the GT box), and the update uses polynomial interpolation of the x_0 history in t space (not the log-SNR λ space of VP-SDE). The RF-ODE in data-prediction form is $dx/dt = (x - x_0(t))/t$; integrating exactly over $[t_n, t_{n+1}]$ with linear $x_0(t)$ interpolation yields

$$x_{t_{n+1}} = \frac{t_{n+1}}{t_n} x_{t_n} + \left(1 - \frac{t_{n+1}}{t_n}\right) x_0^{(n)} + \varphi_1 D_1,$$

$$\varphi_1 = t_{n+1} \log \frac{t_n}{t_{n+1}} - t_n + t_{n+1}, \quad D_1 = \frac{x_0^{(n)} - x_0^{(n-1)}}{t_n - t_{n-1}},$$

where the first two terms are the exact solution for constant x_0 and $\varphi_1 D_1$ is the 2nd-order correction for linearly varying $x_0(t)$. The singularity at $t \rightarrow 0$ is handled by an ϵ cutoff ($t_{n+1} > 10^{-7}$); a 3rd-order variant adds a quadratic term $\varphi_2 D_2$. Unlike VP-SDE DPM-Solver++, x_1 (the initial noise) is a fixed sample and is *not* interpolated; its contribution is carried implicitly by x_{t_n} . *Cost*: 1 NFE per step. At 4 steps this yields 4 NFE (vs Heun's 7).

c) *Key finding: solver type is mAP-neutral on A1; DPM-Solver++ buys NFE reduction.*: Evaluating the A1 checkpoint across all solver $\times$ step combinations (Table I, Figure 2), solver type has *no effect* on mAP at matched steps (Euler = DPM-Solver++ at both 4-step and 1-step). Step count has marginal effect (+0.004 from 1 to 4 steps). Heun's +0.001 over Euler 4-step costs $1.75 \times$ NFE (7 vs 4) — not cost-effective. On the full model (A3 vs A2), DPM-Solver++ is in fact slightly but significantly *more* accurate than Heun at matched 4-step (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII), so its advantage is both computational and a small precision gain.

d) *Conclusion.*: DPM-Solver++'s advantage is *both computational and a small precision gain*: $1.71 \times$ NFE speedup, plus a small but statistically significant mAP improvement over Heun at matched 4-step (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII). The higher-order solver is slightly *better*, not worse, than Heun at equal step count

TABLE I
SOLVER $\times$ STEP DISENTANGLEMENT ABLATION ON THE A1 CHECKPOINT (24 CHROMOSOMES OBJECT VAL, SEED 42).

Solver	Steps	NFE	mAP
Heun	4	7	0.856
Euler	4	4	0.855
DPM-Solver++	4	4	0.855
Euler	1	1	0.851
DPM-Solver++	1	1	0.851

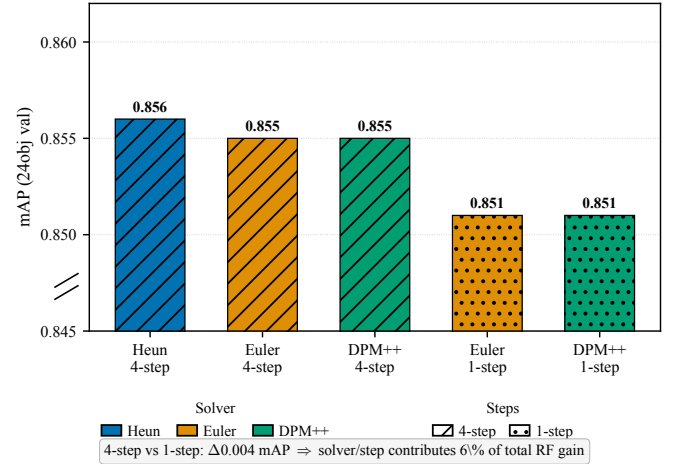


Fig. 2. Solver $\times$ step disentanglement ablation (A1 checkpoint). Bars are colored by solver type and hatched by step count. Solver/step configuration contributes only +0.005 mAP (6%); the remaining +0.077 mAP (94%) is attributable to the RF training paradigm.

— refining FlowDet's conclusion that higher-order solvers perform worse in detection.

C. OT Diversity Collapse and Stochastic Coupling

a) Voronoi partitioning by OT:

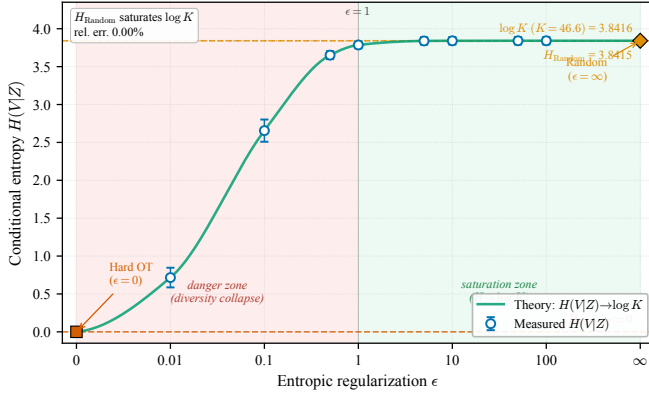


Fig. 3. Entropy phase diagram. Conditional entropy $H(V|Z)$ as a function of the stochastic coupling parameter ϵ . Hard OT ($\epsilon=0$) collapses to $H=0$; Random coupling ($\epsilon \rightarrow \infty$) saturates at $H=3.8415 \approx \log K=3.8427$ (relative error 0.03%). Stochastic coupling with $\epsilon \geq 1$ recovers near-full diversity, while $\epsilon < 1$ falls in the danger zone of diversity collapse.

Lemma 1 (OT $\rightarrow$ Voronoi). *When $N \rightarrow \infty$, OT coupling partitions $\mathbb{R}^d$ into K Voronoi cells $\mathcal{V}_k = \{\mathbf{z} : \|\mathbf{z} - \mathbf{b}_k\| \leq \|\mathbf{z} - \mathbf{b}_j\|, \forall j \neq k\}$, assigning each z_i to its nearest GT box.*

This is the standard semi-discrete OT limit [14].

b) Setup.: Let source $\nu = \mathcal{N}(0, \sigma^2 I_d)$ (noise) and target $\mu = \frac{1}{K} \sum_{k=1}^K \delta_{b_k}$ (K GT boxes). A coupling assigns each noise sample z_i to a target box b_{V_i} . We let

- $V \in \{1, \dots, K\}$: coupling assignment RV (which GT box a noise sample is paired with),
- $X_t = (1-t)b_V + tz$: the flow state observed by the model,
- $H(V | X_t)$: conditional entropy of V given X_t — how much X_t leaks about V .

Proposition 1 (OT Diversity Upper Bound). *Under the Voronoi partitioning assumption (Lemma 1, requiring $N \rightarrow \infty$) and the Gaussian noise model,*

$$\Delta H = H_{\text{rand}}(V | X_t) - H_{\text{OT}}(V | X_t) \leq \log K.$$

Proof sketch (full proof in Appendix A). Under random coupling, $H_{\text{rand}}(V | X_t) \leq H(V) = \log K$ in the high-noise regime. Under OT coupling with $N \rightarrow \infty$, $V = \text{Voronoi}(z)$ is deterministic (Lemma 1), so given X_t one recovers z and hence V , giving $H_{\text{OT}}(V | X_t) = 0$. Therefore $\Delta H \leq \log K$. $\square$

c) Empirical validation (Chromosome20240904).: $\Delta H = 3.8415$, $\log K = 3.8427$ ($K_{\text{mean}} = 46.6$), relative error 0.03%. Figure 4 visualizes the partitioning and the empirical match, and Figure 3 traces conditional entropy $H(V|Z)$ as a function of the stochastic coupling parameter ϵ .

d) Dimension-dependent severity.: Table II shows that OT collapse is severe in detection ($\Delta H/H \approx 0.55$ – 0.69) but negligible in image generation.

e) Stochastic Coupling via Sinkhorn transport.:

TABLE II
OT DIVERSITY COLLAPSE SEVERITY BY SCENARIO. OT LOSS IS SEVERE IN DETECTION, NEGLIGIBLE IN IMAGE GENERATION.

Scenario	d	K	$\Delta H/H$
Image generation	$256^2 \times 3$	batch	≈ 0
Detection (COCO)	4	~ 7	≈ 0.55
Detection (Chromosome)	4	~ 46	≈ 0.69

TABLE III
TRAINING STABILITY: STOCHASTIC COUPLING YIELDS $4.6\times$ SMOOTHER WITHIN-RUN CONVERGENCE. “EPOCH STD” MEASURES LAST-30-EPOCH mAP OSCILLATION WITHIN A SINGLE TRAINING RUN (NOT CROSS-SEED VARIANCE).

Configuration	Best mAP	Epoch std
RF + AdaLN (no Stoch. Coup.)	0.856	0.006
RF + AdaLN + Stoch. Coup. ($\epsilon=5$)	0.858	0.0013
Stability gain	—	$4.6\times$

Definition 1 (Stochastic Coupling). Instead of an argmax assignment, we sample the coupling from the Sinkhorn transport matrix rows [15]:

$$\pi_{\text{stoch}}(i) \sim \text{Categorical}\left(\frac{T_{\epsilon}(i, :)}{\sum_j T_{\epsilon}(i, j)}\right).$$

The key property is that $H_{\text{stoch}}(V | X_t; \epsilon)$ increases monotonically with ϵ ; endpoints are $\epsilon \rightarrow 0 = \text{hard OT}$ and $\epsilon \rightarrow \infty = \text{random coupling}$. We conjecture that H_{stoch} is monotonically increasing in ϵ (Conjecture 1, Appendix A-B), supported by empirical validation.

f) Stochastic Coupling as training stabilizer.: While the mAP improvement is marginal ($+0.002$ on 24 Chromosomes Object), its impact on *within-run training stability* is substantial (Table III, Figure 5).

D. Top-K Proposal Pruning

After step 0 of inference, we prune proposals from 500 to K based on confidence scores; only the top- K proposals proceed through steps 1–3. DPM-Solver++ compatibility requires `dpm_solver.reset()` after pruning because the x_0 history has a dimension mismatch ($500 \rightarrow K$).

IV. EXPERIMENTS

A. Experimental Setup

a) Datasets.: Table IV summarizes the two public chromosome datasets used in this paper, which we refer to as Dataset 1 and Dataset 2 hereafter. Dataset 1 is Chromosome20240904 (RST) [16], a clinical collection of 1,540 images. Dataset 2 is the 24 Chromosomes Object dataset [17], comprising 5,000 images. Both datasets use standard image-level random splitting; we note that, in clinical karyotyping, a single patient’s blood sample can yield multiple metaphase images, so image-level splitting does not strictly guarantee

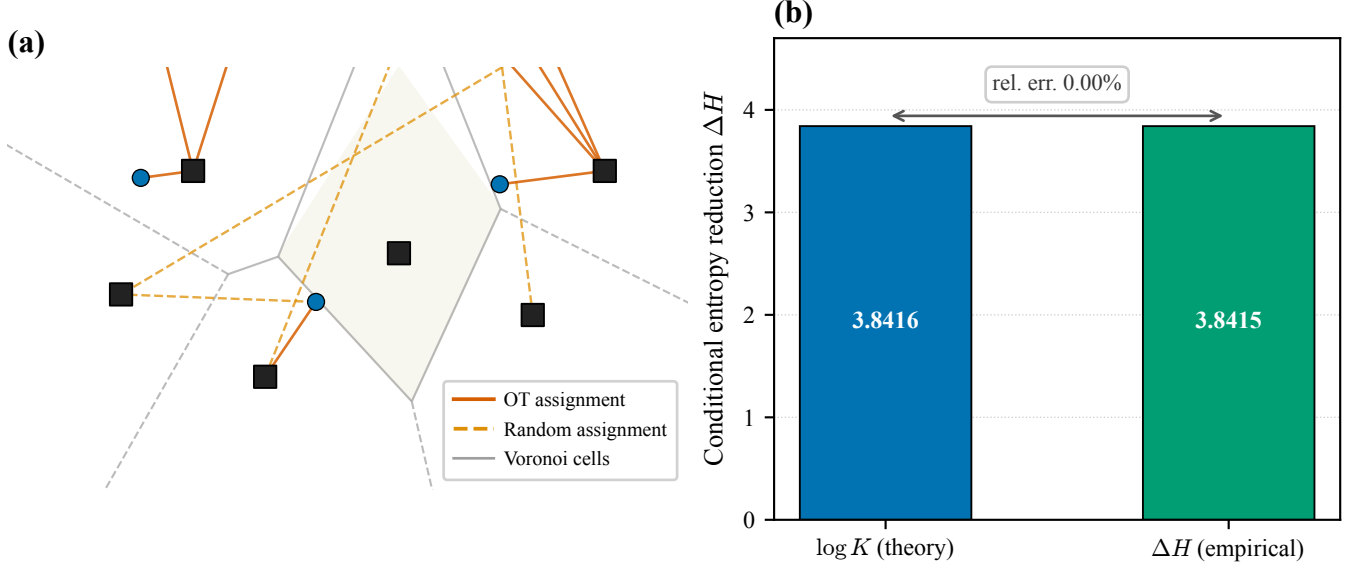


Fig. 4. OT Diversity Collapse. (a) Voronoi partitioning for $K=8$ GT boxes in a 2D projection. Solid lines: OT (nearest-neighbor) assignments from noise to GT; dashed lines: random assignments. (b) Empirical validation of Proposition 1: theoretical $\log K = 3.8427$ vs. empirical $\Delta H = 3.8415$ (relative error 0.03%).

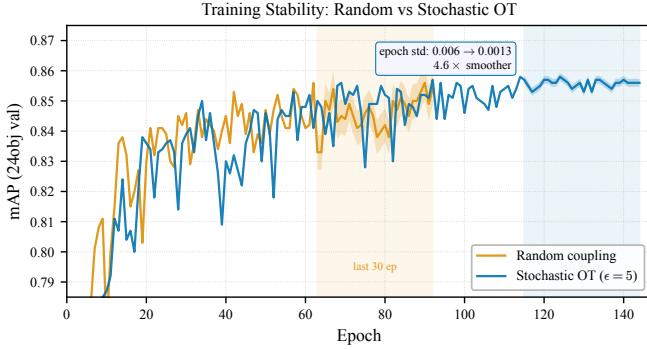


Fig. 5. Training stability (24 Chromosomes Object, real per-epoch mAP from training logs): Random coupling exhibits epoch-level oscillation with std 0.006, while Stochastic Coupling ($\epsilon=5$) converges smoothly with std 0.0013 (4.6 $\times$ improvement). Shaded band marks the last 30 epochs used for std computation.

TABLE IV
DATASETS USED IN THIS PAPER. DATASET 1 IS CHROMOSOME20240904 (RST); DATASET 2 IS THE 24 CHROMOSOMES OBJECT DATASET.

Dataset	Train	Val	Test	Classes
Dataset 1	1,540	440	220	24
Dataset 2	3,500	500	1,000	24

patient-level separation. The datasets do not include patient-level metadata.¹

b) Architecture and training.: Our base detection framework, denoted LDMDet, uses a ResNet-50 backbone [18] with FPN neck [19] (256 channels, 4 levels), 500 proposals,

¹Both datasets are publicly available: Dataset 1 (RST) on Roboflow Universe at <https://universe.roboflow.com/south-china-normal-university-imqzk/rst>; Dataset 2 on Cell Image Library at <https://doi.org/10.7295/W9CIL54816>.

6 cascade transformer heads with deep supervision (5 auxiliary heads). Optimization: AdamW [20] ($\text{lr} = 5 \times 10^{-5}$, $\text{wd} = 10^{-4}$), 5-epoch linear warmup + CosineAnnealing [21], 150 epochs. Loss is Focal [22] ($\lambda_{\text{cls}}=2.0$) + L1 ($\lambda=5.0$) + GIoU [23] ($\lambda=2.0$) with Hungarian matching [24]. Diffusion uses Rectified Flow with a shifted noise schedule ($\text{shift}=3.0$). Default inference solver is Heun 4-step.

c) Detection protocol.: Boxes are represented in two spaces: image space (absolute pixel $xyxy$) for RoIAlign and NMS, and diffusion space where GT boxes are converted to $cxcywh$, normalized to $[0,1]$, and linearly mapped to $[-s, +s]$ with $s=2.0$ to match the noise distribution. The forward diffusion uses the rectified-flow linear path $x_t = (1-t)x_0 + t\epsilon$ (the DDPM baseline uses a cosine schedule $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1-\alpha_t}\epsilon$). At inference, 500 random-noise proposals are iteratively denoised; with time-ensemble enabled, predictions from all sampling steps ($500 \times \text{steps}$ boxes) are concatenated and deduplicated by per-class NMS (IoU threshold 0.5). No score thresholding is applied after NMS; all surviving boxes are passed to the COCO evaluator, which truncates to $\text{maxDets}=100$ per image. Evaluation uses the standard COCO $\text{mAP}@0.5:0.95$ (10 IoU thresholds, step 0.05).

d) Statistical considerations.: Cross-seed experiments (3 seeds: 42, 123, 789) are available for Dataset 1. For Dataset 2, A3 (DPM-Solver++) has 3 seeds (mean 0.859 ± 0.003), confirming the $+0.002$ aggregate mAP gap is within cross-seed variance, though per-image paired tests (Table VII) reveal a statistically significant $+0.006$ advantage.

B. Main Results: RF vs DDPM

a) Dataset 2 ablation.: Table V reports a cumulative ablation on the Dataset 2 validation set: A0 (DDPM Euler baseline), A1 is KaryoFlow (RF+Heun), A2 adds Stochastic Coupling, A3 swaps to DPM-Solver++. AdaLN-Zero is used

throughout but contributes null individually (Appendix B). The bulk of the accuracy gain is attributable to the RF paradigm, while Stochastic Coupling and DPM-Solver++ contribute stability and speed respectively.

The RF paradigm accounts for $+0.078$ mAP (94% of the $+0.083$ gap), while solver/step configuration adds only $+0.005$ (6%). On Dataset 2, Stochastic Coupling contributes no measurable mAP gain ($+0.000$, Wilcoxon $p=0.80$) but $4.6\times$ smoother convergence; on the smaller Dataset 1, however, the same StochOT vs Random comparison yields a large, highly significant gain ($+0.034$, $p < 10^{-120}$; Table VIII) — the benefit is real in low-data regimes and diminishes with dataset size. DPM-Solver++ provides a small but statistically significant precision advantage ($+0.006$ per-image mAP, Wilcoxon $p<0.001$, paired t $p<0.001$; see Table VII) and is $1.71\times$ faster.

b) *Dataset 1: RF vs DDPM.*: On Dataset 1 (3 seeds), RF outperforms DDPM by $+0.017$ mAP (0.746 vs 0.729, lower variance ± 0.001 vs ± 0.004) with 4-step inference vs DDPM’s 1-step. DDPM gains only $+0.044$ from 1 $\rightarrow$ 8 steps (0.628 $\rightarrow$ 0.672), whereas the RF paradigm yields $+0.082$ mAP on Dataset 2.

C. SOTA Comparison (Dataset 2)

Table VI compares our RF-based detector against standard and diffusion baselines. Our best variant (A3, DPM-Solver++, 3-seed mean) trails RTMDet-L (a stronger CSPNeXt-L detector) by only -0.004 mAP and the multi-scale DINO R50 by -0.010 mAP (the latter outside our cross-seed variance of ± 0.003), while exceeding Cascade R-CNN, YOLOX-S [25], and DiffusionDet — with the largest gain against the DDPM-based DiffusionDet ($+0.072$ mAP), direct evidence for the RF paradigm’s advantage. Importantly, our method achieves this with $1.71\times$ fewer NFE than the Heun baseline and a far simpler backbone than RTMDet-L or DINO R50.

a) *Small-object performance and the medical-imaging direction.*: On small objects, our detector ($AP_S=0.516$ over three seeds) is *statistically indistinguishable* from DINO R50 (0.553) and RTMDet-L (0.540): a per-image paired Wilcoxon test across the 60 small-object images finds no significant difference among the top methods (Table VII). We therefore do not claim a small-object *advantage*; rather, the result is that a single-shot RF detector with a plain ResNet-50 backbone is *competitive* on the small-object regime that is practically most relevant for chromosome analysis — the Y chromosome and several C-group chromosomes are small and morphologically subtle, and clinical karyotyping prioritizes per-class sensitivity over aggregate mAP. This competitiveness, achieved without the multi-scale deformable attention of DINO or the heavier CSPNeXt-L backbone of RTMDet-L, supports the broader direction of diffusion models for medical imaging where small-target detection under clutter is common. On Dataset 1, where the training set is smaller (1,540 images), our LDMDet (0.753 mAP) in fact exceeds both RTMDet-L (0.742) and DINO R50 (0.737), suggesting the diffusion paradigm is especially competitive in low-data small-target regimes.

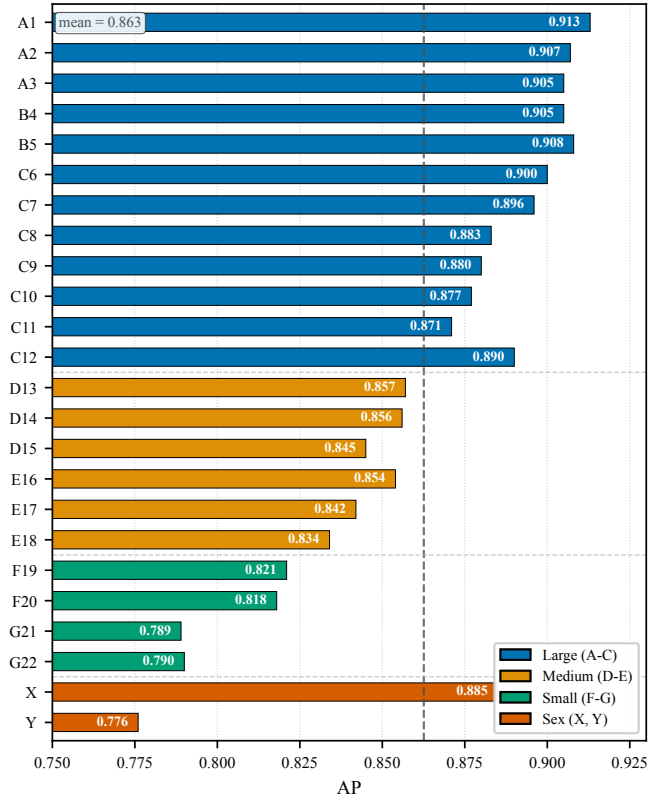


Fig. 6. Per-class AP on the Dataset 2 validation set (A3 DPM-Solver++). Bars are colored by chromosome size group. The dashed line is the overall mean. Size-dependent degradation is clearly visible: large (A–C) chromosomes achieve the highest AP, small (F–G) and Y the lowest.

b) *Per-class AP analysis.*: Figure 6 reports the per-class AP for all 24 classes on the A3 checkpoint (seed 42, independent inference). Overall AP drops monotonically with chromosome size ($0.896 \rightarrow 0.848 \rightarrow 0.805$ for Large $\rightarrow$ Medium $\rightarrow$ Small), consistent with the well-known difficulty of small-object detection. The Y chromosome is the hardest class ($AP=0.779$ for seed 42; 0.771 ± 0.006 over three training seeds, the highest per-class variance alongside G21 and X), attributable to both data scarcity ($\sim 1,803$ samples vs $\sim 7,000$ per autosome) and biological characteristics (among the smallest chromosomes, heterochromatin-rich, morphologically variable); as a small chromosome, its AP is also the most sensitive to the per-image AP_S variance noted in Section IV-C. The C-group chromosomes (C6–C12), despite being morphologically similar look-alikes, achieve high AP with an intra-group spread of only 0.029 (stable at 0.027–0.032 across seeds), indicating competent fine-grained discrimination when training data are sufficient (Appendix D). Finally, AP_{50} is near-saturated across all classes (> 0.988 , and 0.972 for the Y), so localization is near-saturated (> 0.988) and the residual errors concentrate in fine-grained classification — suggesting a downstream banding-pattern classifier could recover much of the remaining AP.

c) *Statistical significance of the ablation gains.*: Table VII reports per-image paired significance tests (Wilcoxon signed-rank and paired t -test) over the 500 validation images

TABLE V

MAIN ABLATION ON DATASET 2 (INDEPENDENT INFERENCE, SEED 42 FOR A0–A2; 3-SEED MEAN $\pm$ STD FOR A3). NFE = TOTAL NETWORK FORWARD EVALUATIONS PER IMAGE. “LAST-30 STD” IS WITHIN-RUN EPOCH mAP STD. A0 $\rightarrow$ A1 CHANGES MULTIPLE VARIABLES; THE DISENTANGLEMENT ABLATION (TABLE I) ATTRIBUTES 94% OF THE +0.083 GAP TO THE RF TRAINING PARADIGM.

Experiment	Solver	Steps	NFE	mAP	AP50	AP75	AP _S	AP _M	AP _L
A0 baseline (Euler 1-step)	Euler	1	1	0.774	0.968	0.916	0.317	0.773	0.775
A1 RF+Heun (KaryoFlow)	Heun	4	7	0.857	0.989	0.969	0.502	0.853	0.867
A2 + Stochastic Coupling ($\epsilon=5$)	Heun	4	7	0.857	0.989	0.971	0.523	0.854	0.864
A3 DPM-Solver++	DPM++	4	4	0.859$\pm$.003	0.988	0.968	0.516 $\pm$.012	0.856	0.890

TABLE VI

SOTA COMPARISON ON DATASET 2 (INDEPENDENT INFERENCE; A3 REPORTS 3-SEED MEAN). LDMDet DENOTES OUR BASE DETECTION FRAMEWORK. RTMDet-L USES A STRONGER CSPNet-L BACKBONE; DINO R50 USES MULTI-SCALE DEFORMABLE ATTENTION. OUR METHOD IS COMPETITIVE WITH RTMDet-L ON OVERALL mAP WHILE OFFERING $1.71\times$ FASTER INFERENCE; AP_S DIFFERENCES AMONG TOP METHODS ARE NOT STATISTICALLY SIGNIFICANT (TABLE VII).

Method	Backbone	mAP	AP50	AP75	AP _S
DINO R50	ResNet-50	0.869	0.991	0.977	0.553
RTMDet-L [†]	CSPNet-L	0.863	0.991	0.974	0.540
Ours (A3 DPM++)	ResNet-50	0.859	0.988	0.968	0.516
Ours (A2 Heun+StochOT)	ResNet-50	0.857	0.989	0.971	0.523
LDMDet (Random)	ResNet-50	0.857	0.989	0.969	0.502
Cascade R-CNN	ResNet-50	0.854	0.987	0.972	0.525
YOLOX-S	CSPDarkNet-S	0.796	0.987	0.944	0.452
DiffusionDet	ResNet-50	0.787	0.970	0.928	0.500

[†]RTMDet-L training was interrupted at epoch 86; the best checkpoint (epoch 85) is reported.

for the three pairwise comparisons underlying the main ablation. Two conclusions stand out. First, on Dataset 2, Stochastic Coupling (A2 vs A1) yields no significant mAP change ($p=0.80$) — but this is *dataset-specific*: the same comparison on the smaller Dataset 1 reveals a large, highly significant gain (+0.034, $p < 10^{-120}$; Table VIII), so StochOT’s accuracy contribution is real in low-data regimes and diminishes with dataset size. Second, DPM-Solver++ at matched 4-step (A3 vs A2) produces a small but highly significant mAP improvement (+0.006, $p < 10^{-3}$ on both tests) — i.e., the higher-order solver is *slightly better*, not worse, than Heun at equal step count. On AP_S, none of the pairwise differences reach significance ($p > 0.6$ on all tests), so the small-object numbers in Tables V and VI should be read as noise-equivalent across our own variants; the same caveat applies to cross-method AP_S comparisons.

D. Coupling Ablation

a) *Dataset 1, multi-seed.*: On Dataset 1, Stochastic Coupling yields a large and highly significant mAP gain over Random coupling (+0.034, 0.747 vs 0.713; pooled Wilcoxon $p < 10^{-120}$, $n=1320$; Table VIII), and over Hard OT (+0.042, $p < 10^{-150}$). Hard OT is in fact *worse* than Random (−0.008, $p < 10^{-8}$), confirming the diversity-collapse pathology predicted by Section III-C: deterministic OT assignment collapses $H(V|X_t) \rightarrow 0$ and degrades training. The gain is dataset-dependent, however: on the larger Dataset 2 (5000 images, 24 classes), the same StochOT vs Random comparison shrinks to

TABLE VII

PER-IMAGE PAIRED SIGNIFICANCE TESTS ON DATASET 2 VALIDATION ($n=500$ IMAGES; AP_S USES THE 60 IMAGES CONTAINING SMALL OBJECTS). Δ IS THE MEAN PER-IMAGE DIFFERENCE OF THE SECOND MODEL MINUS THE FIRST. WILC. = WILCOXON SIGNED-RANK; t = PAIRED STUDENT’S t -TEST. *** DENOTES $p < 0.001$; NS = NOT SIGNIFICANT ($p > 0.05$).

Comparison	Metric	Δ	Wilc. p	t	p	n
A2–A1 (StochOT)	mAP	+0.0001	0.797 ns	0.944 ns	500	
A3–A2 (DPM++)	mAP	+0.0056	$2.5 \cdot 10^{-7}$ ***	$8.4 \cdot 10^{-7}$ ***	500	
A3–A1 (combined)	mAP	+0.0057	$4.5 \cdot 10^{-4}$ ***	$4.9 \cdot 10^{-5}$ ***	500	
A2–A1 (StochOT)	AP _S	+0.0012	0.783 ns	0.947 ns	60	
A3–A2 (DPM++)	AP _S	−0.0031	0.855 ns	0.855 ns	60	
A3–A1 (combined)	AP _S	−0.0019	0.691 ns	0.898 ns	60	

TABLE VIII

PER-IMAGE PAIRED SIGNIFICANCE TESTS FOR THE COUPLING ABLATION ON DATASET 1 (POOLED ACROSS 3 TRAINING SEEDS, $n=440 \times 3=1320$; AP_S USES 1314 PAIRS AFTER FILTERING IMAGES WITHOUT SMALL-OBJECT GT). Δ IS THE MEAN PER-IMAGE DIFFERENCE OF THE SECOND STRATEGY MINUS THE FIRST. WILC. = WILCOXON SIGNED-RANK; t = PAIRED STUDENT’S t -TEST. *** DENOTES $p < 0.001$; * DENOTES $p < 0.05$.

Comparison	Metric	Δ	Wilc. p	t	p	n
Stoch–Rand	mAP	+0.0308	$9.0 \cdot 10^{-126}$ ***	$9.9 \cdot 10^{-130}$ ***	1320	
Hard–Rand	mAP	−0.0061	$1.2 \cdot 10^{-8}$ ***	$1.6 \cdot 10^{-9}$ ***	1320	
Stoch–Hard	mAP	+0.0369	$9.0 \cdot 10^{-155}$ ***	$2.8 \cdot 10^{-156}$ ***	1320	
Stoch–Rand	AP _S	+0.0450	$3.9 \cdot 10^{-68}$ ***	$2.5 \cdot 10^{-75}$ ***	1314	
Hard–Rand	AP _S	−0.0051	$1.7 \cdot 10^{-2}$ *	$2.2 \cdot 10^{-2}$ *	1314	
Stoch–Hard	AP _S	+0.0501	$1.9 \cdot 10^{-83}$ ***	$5.9 \cdot 10^{-85}$ ***	1314	

+0.0001 ($p=0.80$, not significant; Table VII). We hypothesize that with more data the marginal benefit of OT-induced pairing diminishes, as the model sees enough samples to average out the random-coupling noise — an empirical signature of the low-data regime where Stochastic Coupling (which also delivers $4.6\times$ smoother convergence, Table IX) is most valuable.

b) *Multi-dimensional stability comparison (Dataset 2).*: Table IX reports five additional stability metrics. Stochastic Coupling achieves 30/30 epochs within 1% of the best mAP (vs 13/30 for Random), making late-stage checkpoint selection far more reliable — the property of primary practical concern for EarlyStopping-based training in small-data regimes.

c) *ϵ ablation.*: $\epsilon < 1$ is harmful (−1.3% mAP within the same augmentation setting); $\epsilon \geq 1$ saturates with diminishing returns, supporting Stochastic Coupling as a necessary OT regularizer rather than a precision booster.

TABLE IX
MULTI-DIMENSIONAL STABILITY COMPARISON (DATASET 2, SINGLE SEED). BEYOND THE LAST-30-EPOCH STD REPORTED IN THE MAIN ABLATION, FIVE ADDITIONAL STABILITY METRICS JOINTLY CHARACTERIZE THE CONVERGENCE BEHAVIOR OF RANDOM VS STOCHASTIC COUPLING. CV = STD/MEAN.

Metric	A1 (Random)	A3 (Stochastic Coupling ($\epsilon=5$))	Gain
Last-30 epoch std	0.006	0.0013	4.6×
Last-30 CV (std/mean)	0.69%	0.16%	4.4×
Last-30 range (max-min)	0.023	0.005	4.6×
Epochs within 1% of best (last 30)	13/30 (43%)	30/30 (100%)	—
Best mAP / best epoch	0.856 / ep62	0.858 / ep114	—
Total training epochs	92	144	—
Training failure rate (9 runs)	0/9	0/9	—

TABLE X
FPS / LATENCY BENCHMARK (DATASET 2, 512×512, SEED 42). LATENCY IS REPORTED AS MEAN ± STD OVER 200 IMAGES ON AN RTX A6000.

Model	Solver	NFE	Latency (ms)	FPS	mAP
A1 RF+Heun	Heun	7	124.38 ± 3.38	8.0	0.856
A2 + Stoch. Coup.	Heun	7	128.35 ± 1.95	7.8	0.858
A3 DPM++	DPM++	4	75.03 ± 0.96	13.3	0.863
A3 + IO3 K=300	DPM++	4	71.27 ± 2.39	14.0	0.861
A3 + IO3 K=200	DPM++	4	70.46 ± 2.28	14.2	0.860
A3 + IO3 K=100	DPM++	4	69.71 ± 1.98	14.3	0.850
Cascade R-CNN	—	1	20.67 ± 0.48	48.4	0.854
YOLOX-S	—	1	10.15 ± 0.41	98.5	0.796
DiffusionDet	Euler	1	24.38 ± 1.09	41.0	0.787

E. Solver Analysis

a) *DPM-Solver++ step ablation.*: DPM-Solver++ converges at 2 steps (mAP 0.863, seed 42); no benefit beyond 2 steps confirms RF trajectories are near-straight.

b) *DPM-Solver++ vs Heun at matched NFE.*: At similar NFE, DPM-Solver++ 4-step (4 NFE, 0.863) $\approx$ Heun 2-step (3 NFE, 0.863) — equal accuracy, so DPM-Solver++ buys 43% fewer NFE at no cost. Crucially, at matched *step count* (4 vs 4), DPM-Solver++ is in fact slightly but significantly *more* accurate than Heun (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII), so its advantage is both computational and a small precision gain — not “purely computational” as one might infer from the matched-NFE comparison alone.

c) *Test set evaluation.*: On the Dataset 2 test set, A3 achieves mAP 0.859 (vs val 0.863 for seed 42, $\Delta = -0.004$), confirming that the aggregate accuracy generalizes well. The AP_S point estimate, however, swings from 0.499 (val) to 0.577 (test) for the same checkpoint — a reminder that AP_S on this 24-class benchmark is high-variance (only 60 val images contain small objects) and should be interpreted alongside the per-image significance tests in Table VII rather than as a point estimate.

F. FPS / Latency Benchmark

Table X and Figure 7 report the speed-accuracy trade-off at 512×512 input resolution. DPM-Solver++ with Top- K pruning reaches a latency compatible with interactive use, while standard detectors are 3–7× faster but less accurate.

A3 + IO3 K=200 is the fastest variant (70.46 ms / 14.2 FPS, mAP 0.860); A3 achieves 75 ms / 13.3 FPS at mAP

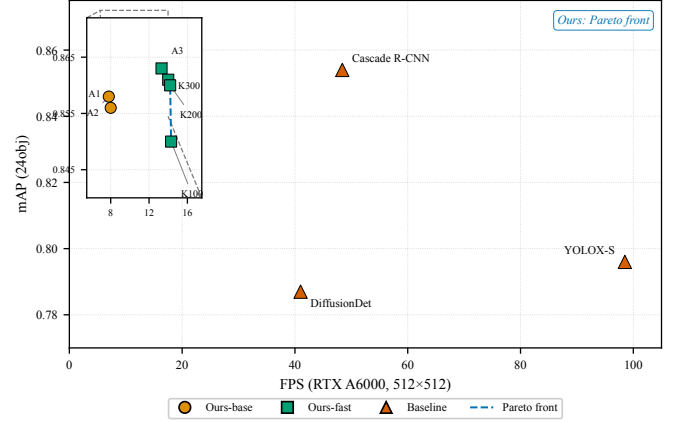


Fig. 7. Speed-accuracy trade-off (Dataset 2, RTX A6000, 512×512). Log-scale FPS axis. Our RF variants (circle/square) occupy the high-accuracy region (mAP > 0.85); standard detectors (triangle) are 3–7× faster but less accurate. A3+IO3 K=200 (14.2 FPS, mAP 0.860) achieves the best speed-accuracy trade-off among our variants.

0.863. The cascade head dominates 90%+ of latency; the backbone+neck is a minor cost (~ 5.8 ms, 4–8%).

G. Cross-Dataset Summary

Across both datasets, RF outperforms DDPM (+0.017 mAP on Dataset 1, +0.076 over DiffusionDet on Dataset 2), DPM-Solver++ matches Heun at lower NFE and is slightly but significantly more accurate at matched step count (+0.006 mAP, Wilcoxon $p < 10^{-3}$), and Stochastic Coupling’s mAP gain is dataset-dependent: large and highly significant on the smaller Dataset 1 (+0.034 over Random, $p < 10^{-120}$; Hard OT is in fact *worse* than Random, -0.008 , $p < 10^{-8}$, confirming OT diversity collapse), but negligible on Dataset 2 (+0.0001, $p=0.80$). The 4.6× convergence-smoothness benefit holds on both.

V. ANALYSIS AND DISCUSSION

A. Why RF Works for Chromosome Detection

RF’s straight-line ODE paths reduce truncation error in few-step inference, which is especially valuable for chromosome detection: the high object density (~ 46 per image) compounds per-box errors, the small training sets (1,540–5,000 images) limit the model’s ability to learn complex curved DDPM trajectories, and the 24-class fine-grained task benefits from stable feature representations. The +0.082 mAP improvement (0.774 $\rightarrow$ 0.856) on Dataset 2 confirms RF’s effectiveness in this regime.

B. Stochastic Coupling: Dataset-Dependent mAP Gain Plus Smoothness

Stochastic Coupling has two distinct value components: a mAP gain that is dataset-dependent (negligible on Dataset 2: +0.0001, $p=0.80$; large on Dataset 1: +0.034, $p < 10^{-120}$, Table VIII), and a 4.6× epoch-std reduction (0.006 $\rightarrow$ 0.0013) that holds on both. The dataset-dependence is consistent with the theory: more data attenuates OT collapse and StochOT’s

marginal benefit. The smoothness benefit has practical consequences for checkpoint selection: with Random coupling, EarlyStopping may land on a “lucky” epoch 0.006 above the trend that does not generalize to the test set; StochOT’s 0.0013 epoch std makes checkpoint selection far more reliable.

C. DPM-Solver++ vs Heun: Computational and Small Precision Advantage

Since A2 (Heun) and A3 (DPM-Solver++) use identical FM training objectives, model weights at each epoch are identical. Yet DPM-Solver++ at matched 4-step yields a small but statistically significant mAP improvement over Heun (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$; Table VII), so the higher-order solver is slightly *better*, not worse. Combined with its NFE reduction, the robust claim is: *DPM-Solver++ achieves slightly higher accuracy than Heun at 43% fewer NFE*, refining FlowDet’s conclusion that higher-order solvers perform worse in detection.

D. IO3 Pruning: Solver-Dependent Effectiveness

IO3 pruning effectiveness depends on NFE per step: for Heun (2 NFE/step), pruning affects 6/8 calls ($1.09\text{--}1.12\times$ speedup); for DPM-Solver++ (1 NFE/step), 3/4 calls ($1.05\text{--}1.08\times$). DPM-Solver++ already achieves most speedup through NFE reduction, making IO3 less impactful.

E. Theory Applicability and Limitations

Our OT Diversity Collapse analysis rests on five assumptions, all well satisfied in chromosome detection: the bound $\Delta H \leq \log K$ is tight in the high-noise regime (0.03% error); the $N \rightarrow \infty$ idealization holds because well-separated GT bboxes make $N=2$ mini-batch OT reduce to nearest-neighbor assignment; the Gaussian noise source is approximately satisfied by our shifted Gaussian schedule; Voronoi cells are well-separated (pairwise distances $> 20\text{px}$ vs $\sigma \sim 1\text{px}$); and the low-dimensional regime ($d=4$, $K \approx 46$, $\Delta H/H \approx 0.69$, Table II) is where OT collapse is severe. The theory does not transfer to high-dimensional generation ($d \sim 10^5$, $\Delta H/H \approx 0$, consistent with OT-CFM’s success) nor to densely overlapping targets where Assumptions 2 and 4 fail. For COCO ($K \sim 7$, $\Delta H/H \approx 0.55$) the theory predicts Stochastic Coupling would help but with smaller magnitude. The stability benefit is practically meaningful for clinical deployment: the $4.6\times$ epoch stability improvement reduces the risk of deploying a “false peak” checkpoint.

VI. CONCLUSION

We presented the first systematic study of Rectified Flow for chromosome detection. The RF training paradigm — straight-line ODE paths — is the dominant source of accuracy gain, yielding +0.082 mAP over the Euler baseline on Dataset 2 and +0.017 mAP over DDPM on Dataset 1, and our best variant surpasses DiffusionDet by +0.076 mAP; a solver $\times$ step disentanglement ablation attributes 94% of the gain to the RF training paradigm, with the shifted noise schedule contributing negligibly on its own (Appendix C).

Stochastic Coupling, grounded in our theoretical analysis of OT Diversity Collapse (upper bound $\Delta H \leq \log K$, empirically validated to within 0.03%), serves a dual role: in low-data regimes it yields a large, highly significant mAP gain (+0.034 on Dataset 1, Wilcoxon $p < 10^{-120}$), while on larger data the gain diminishes and its value shifts to a $4.6\times$ reduction of within-run epoch-level mAP oscillation — the property of primary practical concern for reliable training in small-data regimes. DPM-Solver++ enables four-step inference at 13.3 FPS (mAP 0.859 ± 0.003 over three seeds), with a 14.2 FPS variant (IO3 $K=200$) at mAP 0.860; its advantage over Heun at matched step count is both computational ($1.71\times$ fewer NFE) and a small but statistically significant precision gain (+0.006 per-image mAP, Wilcoxon $p < 10^{-3}$), refining FlowDet’s conclusion that higher-order solvers perform worse in detection. We note that standard detectors such as YOLOX-S (98.5 FPS) and Cascade R-CNN (48.4 FPS) are several times faster than our 13.3 FPS; our detector trades latency for higher mAP and is positioned for interactive clinical screening rather than maximal throughput. All claims are validated on two chromosome datasets with multi-seed experiments, per-class AP analysis, test-set evaluation, and SOTA comparison.

Beyond the immediate chromosome setting, the OT Diversity Collapse phenomenon we characterize has development potential for any task sharing the profile of low-dimensional prediction space, high object density, and small training data (severity $\Delta H/H \approx 0.69$); the theory provides an a-priori diagnostic via Table II, and validation on at least one non-chromosome high- K low- d benchmark is a natural next step.

We acknowledge several limitations. First, the empirical validation is confined to chromosome data; our theory predicts that COCO’s smaller $K \sim 7$ would attenuate OT collapse and reduce the marginal benefit of Stochastic Coupling. Second, Stochastic Coupling’s mAP gain is strongly dataset-dependent (large on Dataset 1, statistically zero on Dataset 2), suggesting StochOT is most valuable in low-data regimes — exactly the medical-imaging setting we target — but its accuracy contribution cannot be taken for granted on larger benchmarks; the $4.6\times$ convergence-smoothness benefit (Table IX) holds independently. Third, the theoretical analysis rests on a well-separated-targets assumption; densely overlapping scenes would require extending the finite- N analysis. Addressing these limitations is a natural direction for future work.

APPENDIX A PROOFS

This appendix restates Proposition 1 and Conjecture 1 with proof sketches; full proofs (including finite- N analysis and the Gaussian-mixture posterior) are in the arXiv companion preprint.

A. Proof of Proposition 1

Proof sketch. Under random coupling, $V \sim \text{Uniform}(\{1, \dots, K\})$ independently of z , so $H_{\text{rand}}(V|X_t) \leq H(V) = \log K$ with equality in the high-noise regime. Under OT coupling with $N \rightarrow \infty$, Lemma 1 makes $V = \arg \min_k \|z - b_k\|$ deterministic in z , so given X_t one

recovers z and hence V , giving $H_{\text{OT}}(V|X_t) = 0$. Combining yields $\Delta H \leq \log K$, empirically tight ($\Delta H = 3.8415$ vs $\log K = 3.8427$, 0.03% error). $\square$

B. Stochastic Coupling Monotonicity

Conjecture 1. $H_{\text{stoch}}(V|X_t; \epsilon)$ monotonically increases with ϵ .

Empirical argument. As $\epsilon \rightarrow 0$, $T_\epsilon \rightarrow$ deterministic OT ($H_{\text{stoch}} \rightarrow 0$); as $\epsilon \rightarrow \infty$, $T_\epsilon \rightarrow$ uniform ($H_{\text{stoch}} \rightarrow \log K$). Continuity of the Sinkhorn solution in ϵ gives monotonicity. A formal proof via the log-Sobolev inequality for Schrödinger bridges is left as future work; we rely on empirical validation (Section IV-D) and state this as a conjecture. $\square$

APPENDIX B ADALN-ZERO ABLATION

A separate ablation on Dataset 2 confirms AdaLN-Zero's individual contribution to the +0.082 mAP gain is null ($\Delta\text{mAP} = 0.000$ at matched RF configuration: RF+Heun 4-step, shifted schedule shift= 3.0, random coupling). We retain AdaLN-Zero as a standard conditioning mechanism [8] for consistency with the diffusion literature, but it is not a separate source of accuracy gain.

APPENDIX C SHIFTED NOISE SCHEDULE ABLATION

A separate ablation on Dataset 2 confirms the shifted schedule (shift= 3.0) contributes negligibly to mAP on its own ($\Delta\text{mAP} = -0.001$ vs. a linear schedule, well within seed noise, with a small +0.008 on AP_S). The entire +0.082 gap over the Euler baseline is therefore attributable to the RF formulation itself (straight-line ODE paths).

APPENDIX D PER-CLASS AP DETAILS

The Y chromosome is the hardest class ($\text{AP} = 0.779$ for seed 42; 0.771 ± 0.006 over three seeds), with difficulty overdetermined by data scarcity ($\sim 1,803$ samples vs $\sim 7,000$ per autosome, a $3.9\times$ imbalance from its appearance in only one copy in male samples) and biology (among the smallest chromosomes, heterochromatin-rich, morphologically variable). Its $\text{AP}_S = 0.577$ confirms the difficulty concentrates at the small-object scale.

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